

Mathematical Analysis II Notes

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8 空间解析几何

Linear Algebra and Analytic Geometry

8.0.1 Inner Product

Let V be a real vector space, an inner product on V is a map.

$$\begin{aligned} g : V \times V &\rightarrow \mathbb{R} \\ (u, w) &\mapsto g(u, w) \end{aligned}$$

satisfying

1. **Bilinearity:**

$$\begin{aligned} g(\lambda_1 v_1 + \lambda_2 v_2, w) &= \lambda_1 g(v_1, w) + \lambda_2 g(v_2, w) \\ g(v, \lambda_1 w_1 + \lambda_2 w_2) &= \lambda_1 g(v, w_1) + \lambda_2 g(v, w_2) \end{aligned}$$

2. **Symmetric:** $g(v, w) = g(w, v), \forall (v, w) \in V$

3. **Positive Definite 正定:** $g(v, v) \geq 0, \forall v, “=” \text{ iff } v = 0$

An inner product is also denoted by $\langle \cdot, \cdot \rangle_g$, or *angle* $\langle \cdot, \cdot \rangle$, or “ $\cdot$ ” if there is no confusion.

Eg. $V = C[0, 1], \langle f, g \rangle \triangleq \int_0^1 f(x)g(x) dx$

For $\dim_{\mathbb{R}} V = n < +\infty$, we have the following classification results:

Thm (Classification of **Symmetric Bilinear Form**). *If $\dim_{\mathbb{R}} V = n$, then for any symmetric bilinear form g on V , there exists a basis $(v_1, \dots, v_n)$ s.t.*

$$(g(v_i, v_j))_{1 \leq i, j \leq n} = \begin{bmatrix} 1 & & & & & & & & \\ & \ddots & & & & & & & \\ & & 1 & & & & & & \\ & & & -1 & & & & & \\ & & & & \ddots & & & & \\ & & & & & -1 & & & \\ & & & & & & 0 & & \\ & & & & & & & \ddots & \\ & & & & & & & & 0 \end{bmatrix} \quad (r_+ \uparrow 1, r_- \uparrow -1, n - r_+ - r_- \uparrow 0)$$

where $r_+ = \max \{ \dim_{\mathbb{R}} W \mid g|_W > 0 \text{ for a subsequence } W \}$, r_- similarly defined.

is called the positive/negative inertia index respectively.

Thm (Classification of Inner Product). *If $\dim_{\mathbb{R}} V = n$, then for any inner product g on V ,*

1. there exists a basis $(v_1, \dots, v_n)$ s.t.

$$g(v_i, v_j) = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}, \text{ i.e. } g(v_i, v_j) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \text{ (**Kronecker Symbol**)}$$

i.e. orthonormal basis w.r.t. g exists.

2. any basis $(v'_1, \dots, v'_n)$ is orthonormal w.r.t. g iff. $(v'_1, \dots, v'_n) = (v_1, \dots, v_n)P$ for some $P \in O(n) = \{A \in M_{n \times n}(\mathbb{R}) \mid AA^T = I\}$, **called the orthonormal group 正交群**

In coordinates

Let V be a n -dim real vector space, an ordered set of n linearly independent vectors is called an ordered basis, usually denoted by $(v_1, \dots, v_n)$.

Fix a particular choice of ordered basis $(v_1, \dots, v_n)$, we have one-to-one correspondence.

1. $V \rightarrow \mathbb{R}^n$ **isomorphism 同构**

$$v \rightarrow x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \text{ coordinate vector (w.r.t. the ordered basis } v_1, \dots, v_n)$$

$$\text{s.t. } v = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

2.

$L(v) = \{\text{linear map from } V \text{ to } V\} \rightarrow M_{n \times n}(\mathbb{R})$ **isomorphism**

$\mathcal{A} \mapsto A$ representing the linear map V under the basis $(v_1, \dots, v_n)$

$$\mathcal{A} \cdot (\mathcal{B} + \mathcal{C}) = \mathcal{A} \cdot \mathcal{B} + \mathcal{A} \cdot \mathcal{C} \quad A(B + C) = AB + AC$$

3.

$\{\text{Bilinear form on } V\} \rightarrow M_{n \times n}(\mathbb{R})$ **isomorphism**

$$g \mapsto (g(v_i, v_j))_{1 \leq i, j \leq n}$$

4.

$\{\text{Symmetric Bilinear form on } V\} \rightarrow \text{Sym}_{n \times n}(\mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}) \mid A^T = A\}$ **isomorphism**

$$g \mapsto (g(v_i, v_j))_{1 \leq i, j \leq n}$$

5. $\{\text{inner product on } V\} \rightarrow \text{Sym}_{n \times n}^+(\mathbb{R}) = \{A \in \text{Sym}_{n \times n}(\mathbb{R}) \mid A > 0\}$

(5) $\subseteq$ (4) **convex cone** $\rightarrow g_1, g_2 \Rightarrow \lambda_1 g_1 + \lambda_2 g_2, \lambda_1 \lambda_2 \geq 0$, but $\lambda_1^2 + \lambda_2^2 \neq 0, \dots$

Thm. For any $A \in \text{Sym}_{n \times n}(\mathbb{R})$, there exists $P \in GL(n, \mathbb{R}) = \{B \in M_{n \times n}(\mathbb{R}) \mid \det B \neq 0\} \rightarrow$
General Linear Group $GL(v)$

$$s.t. PAP^T = \begin{pmatrix} 1 & & & & & & & \\ & \ddots & & & & & & \\ & & 1 & & & & & \\ & & & -1 & & & & \\ & & & & \ddots & & & \\ & & & & & -1 & & \\ & & & & & & 0 & \\ & & & & & & & \ddots \\ & & & & & & & & 0 \end{pmatrix}$$

Thm. For any $A \in \text{Sym}_{n \times n}^+(\mathbb{R})$, there exists $P \in GL(n, \mathbb{R})$, s.t. $PAP^T = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}$

$$(\Rightarrow A = P^{-1} (P^{-1})^T)$$

Thm (Spectral Theorem / Orthogonal Diagonalization). For any $A \in \text{Sym}_{n \times n}(\mathbb{R})$, there exists $P \in O(n)$ and $\lambda_1, \dots, \lambda_n \in \mathbb{R}$, s.t. $PAP^T = \text{diag}(\lambda_1, \dots, \lambda_n) = A$ ($\Rightarrow$ **If $A > 0$, $A = P^T \Lambda P = P^T \Lambda^{\frac{1}{2}} P \cdot P^T \Lambda^{\frac{1}{2}} P = B^2$, where $B^T = B$, $B > 0$**)

8.0.2 Geometric notions induced from inner product g on V

1. Length of a vector: $v \in V, |v|^g = \sqrt{\langle v, v \rangle_g}$
2. Angle: for $v, w \in V \setminus \{0\}$, the angle between v and w is defined as

$$\theta(v, w) = \cos^{-1} \frac{\langle v, w \rangle_g}{|v|_g |w|_g}$$

(where $\cos^{-1}$ is the inverse of $\cos : [0, \pi] \rightarrow [-1, 1]$)

(Cauchy-Schwarz Inequality: $|\langle v, w \rangle_g| \leq |v|_g |w|_g$)

If $\theta(v, w) = \frac{\pi}{2}$, we say v is perpendicular to w w.r.t. g , denoted by $v \perp_g w$, or $v \perp w$ if no confusion.

3. Orthogonal projection: for $\vec{b} \in V \setminus \{0\}$



$$\text{pr}_{\vec{b}} : V \rightarrow \langle \vec{b} \rangle$$

$$\vec{a} \mapsto \frac{\langle \vec{a}, \vec{b} \rangle_g}{|\vec{b}|_g} \frac{\vec{b}}{|\vec{b}|_g}$$

$$\text{pr}_{\langle \vec{b} \rangle^\perp} : V \rightarrow \langle \vec{b} \rangle^{\perp_g} \text{ orthogonal complement}$$

$$\vec{a} \mapsto \vec{a} - \text{pr}_{\vec{b}} \vec{a}$$

$$V = \langle \vec{b} \rangle \oplus \langle \vec{b} \rangle^{\perp_g}$$

4. Volume of “ k -dimensional” parallelepiped 平行多面体 is defined inductively.

The set $P_{v_1, \dots, v_k} = \left\{ \sum_{i=1}^k t_i v_i \mid 0 \leq t_1, \dots, t_n \leq 1 \right\}$ is called a parallelepiped spanned by $(v_1, \dots, v_k)$ in V .

- $k = 1$, $\text{Vol}_g(P_v) = |v|_g$
- $k = 2$, $\text{Vol}_g(P_{v_1, v_2}) \triangleq \text{Vol}_g(P_{v_1}) \left| \text{pr}_{\langle v_1 \rangle^\perp} v_2 \right|_g$

$$\begin{aligned} \text{Vol}_g(P_{v_1, v_2})^2 &= |v_1|_g^2 \left(|v_2|_g^2 - \left(\frac{\langle v_1, v_2 \rangle_g}{|v_2|_g} \right)^2 \right) \\ &= |v_1|_g^2 |v_2|_g^2 - \langle v_1, v_2 \rangle_g^2 \\ &= \det \begin{pmatrix} \langle v_1, v_1 \rangle_g & \langle v_1, v_2 \rangle_g \\ \langle v_2, v_1 \rangle_g & \langle v_2, v_2 \rangle_g \end{pmatrix} \end{aligned}$$

$$v_2 = \text{pr}_{v_1} v_2 + \text{pr}_{v_1^\perp} v_2$$

$$|v_2|_g^2 = |\text{pr}_{v_1} v_2|_g^2 + |\text{pr}_{v_1^\perp} v_2|_g^2$$

$$\text{Thm. } \text{Vol}_g(P_{v_1, \dots, v_k}) = \sqrt{\det \left(\langle v_i, v_j \rangle_g \right)}$$

8.0.3 Orientation 定向

Def (Equivalence Relation). A subset $S \subseteq X \times X = \{(x, y) \mid x \in X, y \in Y\}$ is called a binary relation on X . A binary relation S on X is called an **equivalence relation** 等价关系 on X if

1. Symmetry: $(x, y) \in S \Rightarrow (y, x) \in S$
2. Transitivity: $(x, y) \in S, (y, z) \in S \Rightarrow (x, z) \in S$

3. *Reflexivity*: $(x, x) \in S, \forall x \in X$

It is usually denoted by “ $\sim_S$ ” or “ $\sim$ ”

Def (Equivalence Class). Let “ $\sim$ ” be equivalence relation on X , then for any $x \in X$, the set $\{y \in X \mid y \sim x\} \subseteq X$ is called an equivalence class, it is denoted by $[x]_{\sim}$. Any $y \in [x]_{\sim}$ is called a representative of $[x]_{\sim}$. $X/\sim \triangleq \{[x]_{\sim} \mid x \in X\}$ is called the quotient set of X under the relation “ $\sim$ ”.

Eg. 1. $\{(x, y) \in \mathbb{R}^2 \mid x > y\}$ (“ $>$ ”) is a binary relation on $\mathbb{R}$, but not an equivalence relation.

2. $S = \Delta x = \{(x, x) \mid x \in X\}$ (“ $=$ ”) is an equivalence relation.

3. (Congruence) Fix $n \in \mathbb{N}$

$$\{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid n \mid (x - y)\}$$

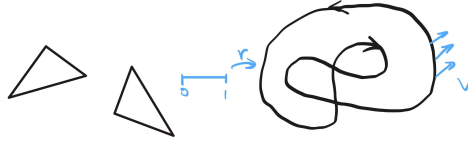
This equivalence relation is called “ $\text{mod } n$ congruence” ($\equiv n$)

$$\mathbb{Z}/\equiv n \left(\triangleq \mathbb{Z}_n \right) = \{[0]_{\equiv n}, \dots, [n-1]_{\equiv n}\}$$

$$\mathbb{Z}_2 = \{[0]_{\equiv 2}, [1]_{\equiv 2}\}$$

$[k]_{\equiv n} = \{l \in \mathbb{Z} \mid n \mid (l - k)\} = \{k + mn \mid m \in \mathbb{Z}\}$, is called the “ $\text{mod } n$ congruence class of k ”.

Eg. 1. $X = \{\text{triangle on } \mathbb{R}^2\} \cong \cong$



2. “Similarity” of matrix $A \sim B \iff A = PBP^{-1}$ for some $P \in GL(n, \mathbb{R})$

$$M_{n \times n}(\mathbb{R})/\sim = \{\text{Jordan canonical form}\}$$

Def (Orientation 定向, Oriented Vector Space 定向线性空间). Let $\mathcal{B}_V = \{\text{Ordered basis of } V\}$ where V is a real vector space, define

$$(v'_1, \dots, v'_n) = (v_1, \dots, v_n)P$$

for some $P \in GL(n, \mathbb{R})$, $\det P > 0$ iff $(v'_1, \dots, v'_n) \sim (v_1, \dots, v_n)$.

This is an equivalence relation, and $B_V/\sim = \{o_1, o_2\}$, an element of $B_V/\sim$ is called an orientation on V . Any vector space, together with a choice of orientation, is called an oriented vector space.

8.0.4 $\mathbb{R}^n$: Euclidean Space

Equipped with the standard inner product

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

where $x = (x_1, \dots, x_n)^+, y = (y_1, \dots, y_n)^+ \in \mathbb{R}^n$.

Under this inner product, $(e_1, \dots, e_n)$, where $e_i = (0, \dots, 0, \overset{i}{1}, 0, \dots, 0)$, is called the *standard orthonormal basis*.

Rmk. $n = 2$, (e_1, e_2) is usually denoted by i, j ;

$n = 3$, (e_1, e_2, e_3) is denoted by i, j, k

Def (Cross Product in $\mathbb{R}^3$). Let $x, y \in \mathbb{R}^3$, define " $\times$ " via

$$\begin{aligned} x \times y &= \begin{vmatrix} i & j & k \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = \begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix} i - \begin{vmatrix} x_1 & x_3 \\ y_1 & y_3 \end{vmatrix} j + \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} k \\ &= (x_2 y_3 - x_3 y_2) i + (x_3 y_1 - x_1 y_3) j + (x_1 y_2 - x_2 y_1) k \end{aligned}$$

call it the cross product of x and y .

Prop. 1. $x \times y = (-y \times x)$ ($x \times y = 0 \iff \{x, y\}$ linear dependent) (*Anti-Symmetric*)

2. $(\lambda x + \mu y) \times z = \lambda x \times z + \mu y \times z$ (*Bilinear*)

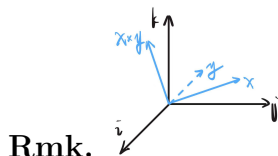
3. $\langle x \times y, x \rangle = 0$ $\langle x \times y, y \rangle = 0$; $(x, y, x \times y)$ determines the same orientation as (i, j, k)

4. $|x \times y| = \text{Area}(P_{x,y})$

Pf. 3.

$$\begin{aligned} (x, y, x \times y) &= (i, j, k) \begin{vmatrix} x_1 & y_1 & x_2 & y_2 \\ x_2 & y_2 & x_3 & y_3 \\ x_3 & y_3 & x_1 & y_1 \\ x_1 & y_1 & x_2 & y_2 \end{vmatrix} \\ &= |x \times y|^2 \end{aligned}$$

$\implies$ if $\{x, y\}$ are linearly independent, $(x, y, x \times y)$ has the same orientation as (i, j, k) .



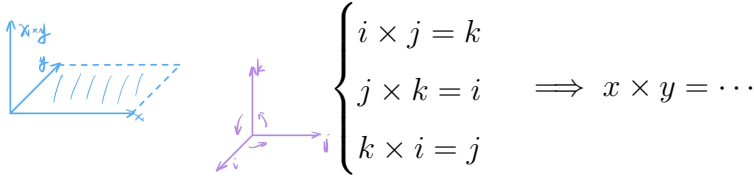
Rmk.

- The orientation denoted by (i, j, k) is called *Right-Hand orientation*;
- The orientation denoted by $(i, j, -k)$ is called *Left-Hand orientation*;
- The ordered basis with the *Right-Hand orientation* is called a *Right-Hand system*.

4.

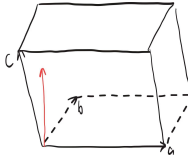
$$\begin{aligned}
 |x \times y|^2 &= (x_2y_3 - x_3y_2)^2 + (x_3y_1 - x_1y_3)^2 + (x_1y_2 - x_2y_1)^2 \\
 \text{Area}(P_{x,y}) &= \det \begin{pmatrix} \langle x, x \rangle & \langle x, y \rangle \\ \langle y, x \rangle & \langle y, y \rangle \end{pmatrix} \\
 &= (x_1^2 + x_2^2 + x_3^2)(y_1^2 + y_2^2 + y_3^2) - (x_1y_1 + x_2y_2 + x_3y_3)^2
 \end{aligned}$$

Thm. In $\mathbb{R}^3$, the cross product is uniquely determined by the four properties.



$$\begin{cases} i \times j = k \\ j \times k = i \\ k \times i = j \end{cases} \implies x \times y = \dots$$

Def (Mixed Product on $\mathbb{R}^3$ 混合积). Let $a, b, c \in \mathbb{R}^3$

$$\langle a, b \times c \rangle = \langle b, a \times c \rangle = \langle c, a \times b \rangle = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$


is called the mixed product of a, b, c .

- $\langle c, a \times b \rangle$ measures the “**signed volume**” of $P_{a,b,c}$;
- This equation also gives the formula for the volume of a parallelepiped in $\mathbb{R}^3$;
- $\langle c, a \times b \rangle \begin{cases} > 0 & (a, b, c) \text{ is a Right-Hand system} \\ = 0 & \{a, b, c\} \text{ linearly dependent} \\ < 0 & (a, b, c) \text{ is a Left-Hand system} \end{cases}$

Recall:

$$\text{Vol}(P_{v_1, \dots, v_k}) = \sqrt{\det(\langle v_i, v_j \rangle)}, \quad v_1, \dots, v_k \in \mathbb{R}^n$$

If $v_i = \sum_{s=1}^n e_s P_{s_i}$ ($\iff (v_1, \dots, v_k) = (e_1, \dots, e_n) P$), then

$$\begin{aligned}
 \langle v_i, v_j \rangle &= \left\langle \sum_{s=1}^n e_s P_{s_i}, \sum_{t=1}^n e_t P_{t_j} \right\rangle = \sum_{s,t=1}^n P_{s_i} P_{t_j} \delta_{st} = \sum_{s=1}^n P_{s_i} P_{s_j} \\
 &= (P^T P)_{ij}
 \end{aligned}$$

$$\implies \text{Vol}(P_{v_1, \dots, v_k}) = \sqrt{\det(P^T P)}$$

$$(\implies \text{In } \mathbb{R}_3, \text{Vol}(P_{a,b,c}) = \sqrt{\det(P^T P)} = \sqrt{(\det P)^2} = |\det P|)$$

Thm (Cauchy-Binet).

$$\det(U^T U) = \sum_{\substack{1 \leq i_1 < \dots < i_k \leq n \\ \binom{n}{k}}} U \begin{pmatrix} i_1 & \dots & i_k \\ 1 & \dots & k \end{pmatrix}^2$$

$$U^T U = \begin{pmatrix} U_{11} & \dots & U_{n1} \\ \vdots & & \vdots \\ U_{1k} & \dots & U_{nk} \\ \uparrow \\ i_1 & i_k \end{pmatrix} \begin{pmatrix} U_{11} & \dots & U_{1k} \\ \vdots & & \vdots & \leftarrow i_1 \ i_k \\ U_{n1} & \dots & U_{nk} \end{pmatrix}$$

Corollary (Generalized Pythagorean Theorem).

$$\text{Vol}(P_{v_1, \dots, v_k})^2 = \sum_{1 \leq i_1 < \dots < i_k \leq n} U \begin{pmatrix} i_1 & \dots & i_k \\ 1 & \dots & k \end{pmatrix}^2 \quad (k = 1, |V|^2 = \langle V, e_1 \rangle^2 + \dots + \langle V, e_n \rangle^2)$$

8.1 向量与坐标系

8.1.1 向量的定义与向量的加法和数乘

8.1.2 向量的共线和共面

8.1.3 向量的点乘和叉乘

8.1.4 向量的坐标表示

8.1.5 空间坐标系

8.2 Lines and Planes in $\mathbb{R}^3$

8.2.1 Plane

2-dim'l affine subspace.

- (General Equation) $\Pi : ax + by + cz + d = 0$ **Implicit equation**

Suppose $r_0 = (x_0, y_0, z_0) \in \Pi$, i.e. $ax_0 + by_0 + cz_0 + d = 0$

$$\begin{aligned} ax + by + cz + d = 0 &\iff a(x - x_0) + b(y - y_0) + c(z - z_0) = 0 \\ &\iff \langle \vec{r} - \vec{r}_0, \vec{n} \rangle = 0 \text{ where } \vec{n} = (a, b, c)^T \\ &\quad \text{(Point-Norm Form 点法式)} \end{aligned}$$

- (Point-Directions Form) Take $\vec{r}_0 \in \Pi$, $Q, R \in \Pi$, s.t. $\overrightarrow{PQ}, \overrightarrow{PR}$ linearly independent,
 $\vec{u} = \overrightarrow{PQ}$, $\vec{v} = \overrightarrow{PR}$, then Π can be explicitly parameterized by

$$\vec{r} = \vec{r}_0 + s\vec{u} + t\vec{v}$$

$$\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR} \text{ normal vector to } \Pi$$

8.2.2 Line

1-dim'l affine subspace, i.e. shift of 1-dim'l vector subspace.

- (Point-Direction Form) $\vec{r} = \vec{r}_0 + t\vec{v}$, $t \in \mathbb{R}$ **Explicit equation 显式表达**
- (Plane-Intersection) The set of $\mathbb{R}^3$ defined by $\begin{cases} a_1x + b_1y + c_1z + d_1 = 0 \\ a_2x + b_2y + c_2z + d_2 = 0 \end{cases}$, satisfying
 $\text{rk} \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{pmatrix} = 2$. **Implicit equation 隐式表达**

- (1) $\implies$ (2) $(\vec{r} - \vec{r}_0) \times \vec{v} = \vec{0}$

$$\begin{aligned} \begin{pmatrix} i & j & k \\ x - x_0 & y - y_0 & z - z_0 \\ l & m & n \end{pmatrix} = 0 &\implies \begin{cases} n(y - y_0) = m(z - z_0) \\ l(z - z_0) = n(x - x_0) \\ m(x - x_0) = l(y - y_0) \end{cases} \\ &\implies \left(\begin{array}{ccc|c} 0 & n & -m & * \\ -n & 0 & l & * \\ m & l & 0 & * \end{array} \right) \end{aligned}$$

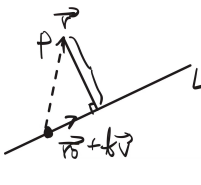
- (2) $\implies$ (1) $\xrightarrow{\text{Particular solution}} \vec{r}_0 + t(\vec{n}_1 \times \vec{n}_2)$ where $\vec{n}_1 = (a_1, b_1, c_1)^T$, $\vec{n}_2 = (a_2, b_2, c_2)^T$

8.2.3 Relation between Lines & Planes

$$1. \text{ Two planes: } \begin{cases} \Pi_1 : a_1x + b_1y + c_1z + d_1 = 0 & \rightsquigarrow \vec{n}_1 \\ \Pi_2 : a_2x + b_2y + c_2z & \rightsquigarrow \vec{n}_2 \end{cases}$$

$$(a) \text{ Parallel: } \vec{n}_1 \parallel \vec{n}_2$$

$$(b) \text{ Intersecting: } \theta = \arccos \frac{\langle \vec{n}_1, \vec{n}_2 \rangle}{|\vec{n}_1| |\vec{n}_2|} ,$$

$$2. \text{ Point \& line: } d(P, L) = \frac{|(\vec{r} - \vec{r}_0) \times \vec{v}|}{|\vec{v}|}$$


$$\text{Point \& plane: } \Pi : ax + by + cz + d = 0 \implies d(P, \Pi) = \frac{|ax + by + cz + d|}{\sqrt{a^2 + b^2 + c^2}}$$

$$P = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

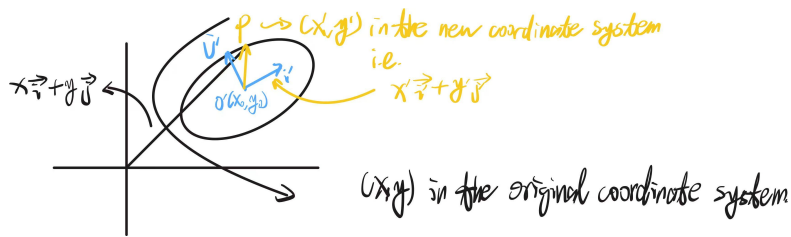
$$d(P, \Pi) = (\vec{r} - \vec{r}_0) \cdot \vec{n} = \frac{(a(x-x_0) + b(y-y_0) + c(z-z_0))}{\sqrt{a^2 + b^2 + c^2}}$$

$$3. \text{ Two lines: } \begin{cases} L_1 : \vec{r} = \vec{r}_1 + t\vec{v}_1 \\ L_2 : \vec{r} = \vec{r}_2 + t\vec{v}_2 \end{cases}$$

$$(a) \text{ Co-planar } \iff \langle \vec{r}_1 - \vec{r}_2, \vec{v}_1 \times \vec{v}_2 \rangle = 0$$

$$(b) \text{ Distance: } \left\langle \vec{r}_2 - \vec{r}_1, \frac{\vec{v}_1 \times \vec{v}_2}{|\vec{v}_1 \times \vec{v}_2|} \right\rangle$$

8.3 Conic Section/Curve & Quadric Surface



$$\Rightarrow \begin{pmatrix} \vec{i} & \vec{j} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \overrightarrow{OP} = \overrightarrow{OO'} + \overrightarrow{O'P} \\ = \begin{pmatrix} \vec{i}' & \vec{j}' \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} \vec{i}' & \vec{j}' \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix} \\ = \begin{pmatrix} \vec{i}' & \vec{j}' \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = P \begin{pmatrix} x' \\ y' \end{pmatrix} + \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} \text{ coordinate transformation}$$

Quadric in $\mathbb{R}^n$

$$f(x_1, \dots, x_n) = \sum_{i,j=1}^n x_i A_{ij} x_j + \sum_{i=1}^n b_i x_i + C, \text{ where } A = A^T \\ = X^T A X + b^T X + C, \text{ where } X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

Question: Can we find coordinate transformation $x = Py$ where $P \in O(n)$ to simplify the expression:

$$y^T P^T A P y + b^T P y + C$$

1. Take P s.t.

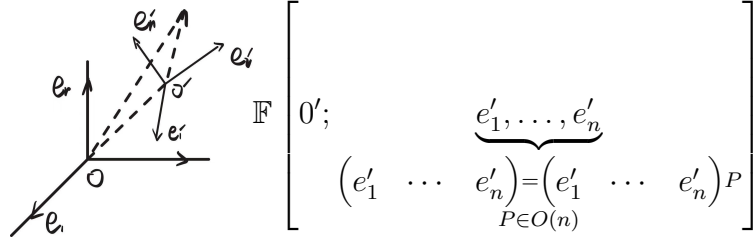
$$P^T A P = \begin{pmatrix} \lambda_1 & & & & & & & \\ & \ddots & & & & & & \\ & & \lambda_{r_+} & & & & & \\ & & & -\lambda_{r_++1} & & & & \\ & & & & \ddots & & & \\ & & & & & \lambda_{r_++r_-} & & \\ & & & & & & 0 & \\ & & & & & & & \ddots \\ & & & & & & & & 0 \end{pmatrix}$$

2. $y = z + \begin{pmatrix} \vdots \end{pmatrix}$

$$3. \lambda_1 z_1^2 + \cdots + \lambda_{r_+} z_{r_+}^2 - \left(\lambda_{r_++1} z_{r_++1}^2 + \cdots \right)$$

$$\text{Quadric surface: } \left\{ \vec{x} \in \mathbb{R}^n \mid f(\vec{x}) = \vec{x}^T A \vec{x} + \vec{b}^T \vec{x} + c = 1 \right\}, A = A^T$$

Descartes / Catesian Coordinate



A point has coordinate

$$\left\{ \begin{array}{l} \vec{x}' = \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix} \text{ in } \mathbb{F}[0'; e'_1, \dots, e'_n] \\ \vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \text{ in } \mathbb{F}[0; e_1, \dots, e_n] \end{array} \right. \Rightarrow \begin{array}{l} \vec{OO'} \\ =_{(e_1 \dots e_n)} \vec{v} \end{array} + \begin{pmatrix} e'_1 & \dots & e'_n \end{pmatrix} \vec{x}' = \begin{pmatrix} e_1 & \dots & e_n \end{pmatrix} \vec{x}$$

$$\Rightarrow \vec{x} = \vec{v} + p \vec{x}'$$

Step 1. $\vec{x} = p \vec{x}' \Rightarrow x' P^T A P x' + b^T P x' + c = 0$
 对角阵

Find P s.t.

$$\xrightarrow{\text{orthogonal diagonalization}} P^T A P = \Lambda = \text{diag} \left(\lambda_1, \dots, \lambda_{r_+}, -\lambda_{r_++1}, \dots, -\lambda_{r_++r_-}, \underbrace{0, \dots, 0}_{n-\text{rank } A} \right)$$

$$\text{then } \lambda_1 x_1'^2 + \dots + \lambda_{r_+} x_{r_+}'^2 - \lambda_{r_++1} x_{r_++1}'^2 - \dots - \lambda_{r_++r_-} x_{r_++r_-}'^2 + \mathbf{b}'_1 \mathbf{x}'_1 + \dots + \mathbf{b}'_n \mathbf{x}'_n + c = 0$$

Step 2. Completing square for the variables $x'_1, \dots, x'_{r_+}, x'_{r_++1}, \dots, x'_{r_++r_-}$

$$\begin{pmatrix} x'_1 \\ \vdots \\ x'_{r_+} \\ x'_{r_++1} \\ \vdots \\ x'_{r_++r_-} \\ \vdots \\ x'_n \end{pmatrix} = \begin{pmatrix} x''_1 - \frac{b'_1}{2\lambda_1} \\ \vdots \\ x''_{r_+} - \frac{b'_{r_+}}{2\lambda_{r_+}} \\ x''_{r_++1} - \frac{b'_{r_++1}}{2\lambda_{r_++1}} \\ \vdots \\ x''_{r_++r_-} - \frac{b'_{r_++r_-}}{2\lambda_{r_++r_-}} \\ x''_{r_++r_-+1} \\ \vdots \\ x''_n \end{pmatrix}$$

$$\text{i.e. } x'_1 = x'' + b''$$

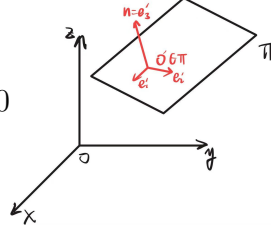
$$\lambda_1 x_1''^2 + \dots + \lambda_{r_+} x_{r_+}''^2 - \lambda_{r_++1} x_{r_++1}''^2 - \dots - \lambda_{r_++r_-} x_{r_++r_-}''^2 + b'_{r_++r_-+1} x_{r_++r_-+1}'' + \dots + b'_n x_n'' + c = 0$$

Step 3. Case 1.

$$b'_{r+1} = \dots = b'_n = 0 \rightsquigarrow x''^T \text{diag}(\lambda_1, \dots, \lambda_{r_+}, -\lambda_{r_++1}, \dots, -\lambda_{r_++r_-}, 0, \dots, 0) x'' + c = 0$$

Case 2. We can do another orthogonal transformation on the variable $x''_{r+1}, \dots, x''_n$ to simplify the 1st-order term s.t. the 1st order term becomes $b'''_{r+1} x'''_{r+1}$

Back to plane in $\mathbb{R}^3$: $ax + by + cz + d = 0$



Let $e'_3 = n = \frac{(a,b,c)}{\sqrt{a^2+b^2+c^2}}$, e'_1, e'_2 parallel to Π s.t. (e'_1, e'_2, n) is an orthonormal basis, then in the Cartesian coordinate $\mathbb{F}[0'; e'_1, e'_2, e'_3]$, Π is defined by $z' = 0$

$$\begin{pmatrix} a & b & c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} + d = 0$$

Let $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = P \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} + q$ (Euclidean transformation), then the equation is changed to

$$\begin{pmatrix} a & b & c \end{pmatrix} P \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} + \begin{pmatrix} a & b & c \end{pmatrix} q + d = 0$$

find $P \in O(3)$, $q \in \mathbb{R}^3$ s.t. $\begin{pmatrix} a & b & c \end{pmatrix} P = \begin{pmatrix} 0 & 0 & c \end{pmatrix}$, $c \neq 0$, $\begin{pmatrix} a & b & c \end{pmatrix} q + d = 0$

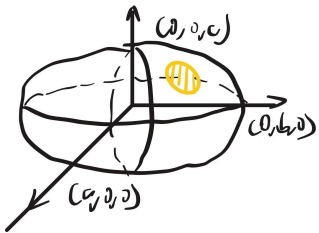
Thm. Given $S = \left\{ \vec{x} \in \mathbb{R}^n \mid f(\vec{x}) = \vec{x}^T A \vec{x} + \vec{b}^T \vec{x} + c = 0 \right\}$, we can find a coordinate transformation of the form $\vec{x} = P \vec{y} + \vec{d}$, where $P \in SO(n) = \{Q \in O(n) \mid \det Q = 1\}$ and $\vec{d} \in \mathbb{R}^n$, such that the equation $f(\vec{x}) = 0$ becomes

$$(1) \sum_{i=1}^{r_+} \lambda_i y_i^2 - \sum_{j=r_++1}^{r_++r_-} \lambda_j y_j^2 = c \text{ or}$$

$$(2) \sum_{i=1}^{r_+} \lambda_i y_i^2 - \sum_{j=r_++1}^{r_++r_-} \lambda_j y_j^2 - y_{r_++r_-+1} = 0$$

Classification of Quadric Surface in $\mathbb{R}^3$

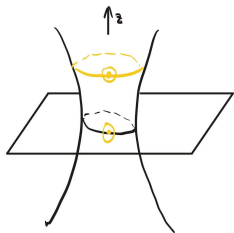
1. rank $A = 3$

$$(+, +, +; +) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$


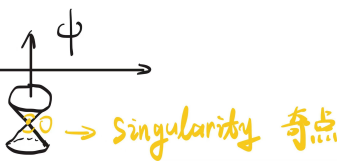
ellipsoid

$$(+, +, +; 0) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 0$$

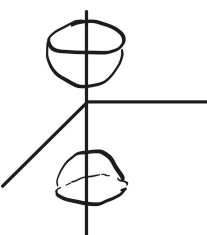
$$(+, +, +; -) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = -1$$

$$(+, +, -; +) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$


single sheeted hyperboloid 单叶双曲面

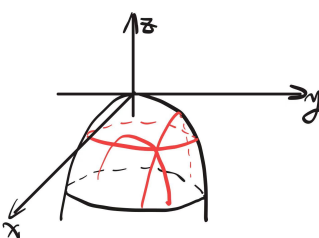
$$(+, +, -; 0) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$$


Singularity 奇点

$$(+, +, -; -) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$$


Double sheeted hyperboloid 双叶双曲面

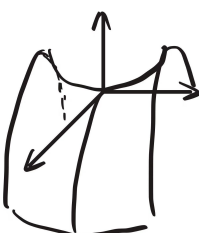
2. rank $A = 2$

$$(+, +, 0; +; 0) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} + z = 0$$


Elliptical paraboloid 椭圆抛物面

$$(+, +, 0; 0; +) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

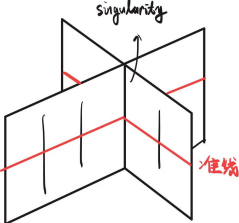

Elliptical cylinder 椭圆柱面

$$(+, -, 0; +; 0) \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} + z = 0$$


Hyperbolic paraboloid/
Saddle surface 鞍面

$$\begin{cases} (+, -, 0; 0; +) & \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \\ (+, -, 0; 0; -) & \frac{x^2}{a^2} - \frac{y^2}{b^2} = -1 \end{cases}$$

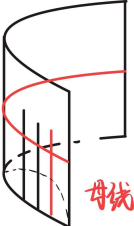

Hyperbolic cylinder

$$(+, -, 0; -; 0) \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$


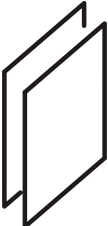
singularity

准线

3. rank $A = 1$

$$(+, 0, 0; +; 0) \quad \frac{x^2}{a^2} + y = 0$$


母线

$$(+, 0, 0; 0; +)$$


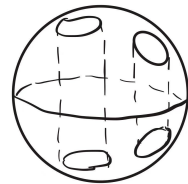
Double plane

$$(+, 0, 0; 0; 0)$$

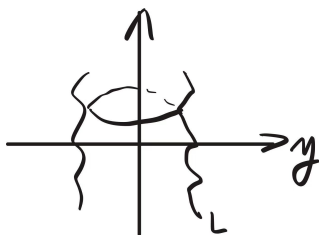
$$(+, 0, 0; 0; -) \rightarrow \emptyset$$

Intersection of Quadric Surface and Plane

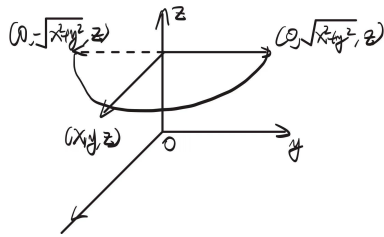
$$\begin{cases} f(\vec{x}) = \vec{x}^T A \vec{x} + \vec{b}^T \vec{x} + c = 0 \\ \vec{n}^T \vec{x} + d = 0 \end{cases} \quad \begin{cases} x^2 + y^2 + z^2 = 1 \\ (x - a)^2 + y^2 = r^2 \end{cases}$$



Surface of Revolution 旋转



$F(y, z) = 0 \rightarrow$ curve L revolved along Z axis $\Rightarrow$ Surface implicit equation?

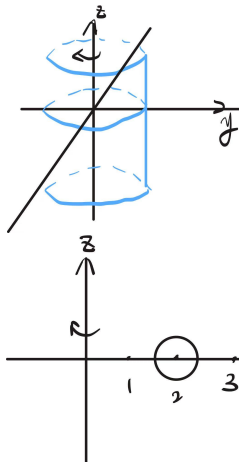


$$\Rightarrow F\left(\pm\sqrt{x^2+y^2}, z\right)=0$$

$\Leftrightarrow (x, y, z)$ lies on the surface of revolution

Eg.

$$L=\left\{(0, y, z) \mid y=z\right\}$$

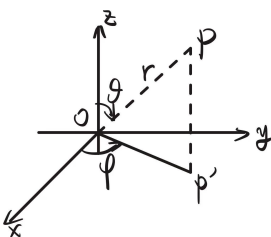


$$\begin{aligned} \pm\sqrt{x^2+y^2}=z &\rightsquigarrow x^2+y^2-z^2=0 \\ x^2+y^2+4\pm 4\sqrt{x^2+y^2}+z^2=1 &\Rightarrow\left(x^2+y^2+z^2+3\right)^2=16\left(x^2+y^2\right) \\ (y-2)^2+z^2=1 &\rightsquigarrow\left(\pm\sqrt{x^2+y^2}-2\right)^2+z^2=1 \end{aligned}$$

8.4 坐标变换和其他常用坐标系

8.4.1 坐标变换

8.4.2 Other Coordinate System

1.  $(r, \theta, \varphi) \begin{cases} r \in [0, +\infty) \\ \theta \in [0, \pi] \\ \varphi \in [0, 2\pi) \end{cases} \rightsquigarrow (x, y, z)^T = (r \sin \theta \cos \varphi, r \sin \theta \sin \varphi, r \cos \theta)$

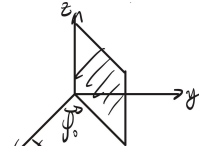
1:1 on the “open point”

$$x^2 + y^2 = r^2 \sin^2 \theta \quad z = r \cos \theta$$

$$\begin{cases} r = \sqrt{x^2 + y^2 + z^2} \\ \theta = \arctan \frac{\sqrt{x^2 + y^2}}{z} \\ \varphi = \arctan \frac{y}{x} \end{cases}$$

- $r = r_0$ 

- $\theta = \theta_0$ 

- $\varphi = \varphi_0$ 

Curvilinear coordinate system

2. Cylindrical coordinate (r, θ, z)

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases} \iff \begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \arctan \frac{y}{x} \\ z = z \end{cases}$$

9 多变量函数的微分学

9.1 多变量函数及其连续性

9.1.1 Point-set

Point-set $\mathbb{R}, \mathbb{R}^n, n \geq 2$

Recall: Distance on $\mathbb{R}^n$ (Euclidean distance)

$$d: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R} \\ (x, y) \mapsto |x - y|$$

- Nonnegativity: $d(x, y) \geq 0, \forall x, y \in \mathbb{R}^n$; “=” is achieved iff $x = y$
- Symmetric: $d(x, y) = d(y, x)$
- Triangle inequality: $d(x, y) \leq d(x, z) + d(z, y), \forall x, y, z \in \mathbb{R}^n$

$$|u + v| \leq |u| + |v| \iff \langle u + v, u + v \rangle \leq \langle u, u \rangle + 2|u||v| + \langle v, v \rangle \\ \iff \langle u, v \rangle \leq |u||v| \text{ (Cauchy-Schwarz inequality)}$$

Def. The disc $B(M_0, r) = \{M \in \mathbb{R}^n \mid d(M, M_0) < r\}$ is called a disc-neighborhood of M_0 , which is also denoted by $B_r(M_0)$. $B^-(M_0, r) = B(M_0, r) \setminus \{M_0\}$ is called punctured disc.

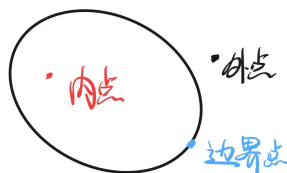
Def. $E \subseteq \mathbb{R}^n$ is called bounded if $E \subseteq B(0, R)$ for some $R > 0$, and the number

$$\text{diam } E \triangleq \sup \{d(M', M'') \mid M', M'' \in E\}$$

is called the diameter of E .

Def. Let $E \subseteq \mathbb{R}^n$ and $E^C = \mathbb{R}^n \setminus E$ be its complement.

1. If $\exists r > 0$, s.t. $B(M, r) \subseteq E$, then M is called an interior point 内点 of E .
2. If $\exists r > 0$, s.t. $B(M, r) \subseteq E^C$, then M is called an exterior point 外点 of E .
3. If $\exists r > 0$, $B(M, r) \cap E \neq \emptyset, B(M, r) \cap E^C \neq \emptyset$, then M is called a boundary point 边界点 of E .



4. $\overset{\circ}{E} = \{M \in \mathbb{R}^n \mid M \text{ is an interior point of } E\}$, is called the interior 内部 of E .
5. **∂ : partial** $\partial E = \{M \in \mathbb{R}^n \mid M \text{ is a boundary point of } E\}$, is called the boundary of E .
6. If $\exists r > 0$, s.t. $B(M, r) \cap E = \{M\}$, then M is called an isolated point 孤立点.
7. If $\forall r > 0$, $B^-(M, r) \cap E \neq \emptyset$, then M is called a accumulation point 聚点.
 $\overline{E} = E \cup \partial E$, is called the closure 闭包 of E .

Rmk. 1. Interior point implies accumulation point

(a) $M \in E \implies M$ is an accumulation point

(b) $M \notin E \implies$ any punctured disc neigh. contains points in $E \implies$ accumulation

2. boundary point $\implies$ either isolated point or accumulation point

boundary point + non-isolated $\implies$ accumulation point

Def. Let $\{M_n\}$ be a sequence of points on $\mathbb{R}^2$, and $M_0 \in \mathbb{R}^2$. If $\lim_{n \rightarrow +\infty} d(M_n, M_0) = 0$, we say M_n converges to M_0 (as $n \rightarrow +\infty$), also denoted by $\lim_{n \rightarrow +\infty} M_n = M_0$

$M_n = \begin{pmatrix} x_n \\ y_n \end{pmatrix}$, $M_0 = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$, then $\lim_{n \rightarrow +\infty} M_n = M_0 \iff \lim_{n \rightarrow +\infty} x_n = x_0$, and $\lim_{n \rightarrow +\infty} y_n = y_0$
i.e. $\sqrt{(x_n - x_0)^2 + (y_n - y_0)^2} \rightarrow 0$, $|x_n - x_0|, |y_n - y_0| \leq d(M_n, M_0) \leq |x_n - x_0| + |y_n - y_0|$

A sequence of points $\{M_n\}$ is called a Cauchy Seq. if

$$\forall \varepsilon > 0, \exists N = N(\varepsilon) \in \mathbb{N}_+, \text{ s.t. } \forall n, m > N, d(M_n, M_m) < \varepsilon$$

Thm. 1. (Completeness) All Cauchy seq. in $\mathbb{R}^n$ has convergent subseq.

2. (Bolzano-Weierstrass Thm.) All bounded seq. in $\mathbb{R}^n$ has convergent subseq.

Def (Open set / Closed set). If $E = \overset{\circ}{E}$, we say E is open. If E^C is open, we say E is closed.

Eg. Disc-neighborhood $B(M, r) = \{O \in \mathbb{R}^n \mid d(P, M) < r\}$ is open.

$$\forall P \in B(M, r), d(P, M) < r$$

$$\implies B(P, r - d(P, M)) \subseteq B(M, r)$$

$$(d(Q, M) \leq d(Q, P) + d(P, M) < (r - d(P, M)) + d(P, M)) = r, \forall Q \in B(P, r - d(P, M))$$

$$\implies B(P, r) \text{ is open, } B(P, r)^C \text{ closed.}$$

$$\partial B(M, r) = \left\{ P \in \mathbb{R}^n \mid \boxed{d(P, M) = r} \right\} = S(M, r) \leftarrow \text{Sphere}$$

Prop. 1. E_1, E_2 open $\implies E_1 \cap E_2$ is open.

(If $E_1, \dots, E_n$ open, then $E_1 \cap \dots \cap E_n$ is open)

If $\{E_\alpha\}_{\alpha \in \Lambda}$ is a family 族 of open sets, then $\bigcup_{\alpha \in \Lambda} E_\alpha$ is open.

$$\begin{aligned} \Longleftarrow \forall x \in \bigcup_{\alpha \in \Lambda} E_\alpha &\implies x \in E_\alpha \text{ for some } \alpha \in \Lambda \\ &\implies \exists r > 0 \text{ s.t. } B(x, r) \subseteq E_\alpha \\ &\implies B(x, r) \subseteq \bigcup_{\alpha \in \Lambda} E_\alpha \end{aligned}$$

$$\bigcap_{n=1}^{\infty} B\left(0, \frac{1}{n}\right) = \{0\} \text{ not open}$$

2. E_1, E_2 closed $\implies E_1 \cup E_2$ is closed.

(If $E_1, \dots, E_n$ closed, then $E_1 \cup \dots \cup E_n$ is closed)

If $\{E_\alpha\}_{\alpha \in \Lambda}$ is a family of closed sets, then $\bigcap_{\alpha \in \Lambda} E_\alpha$ is closed.

3. E is open iff $E \cap \partial E = \emptyset$, i.e. $\partial E \subseteq E^C$

4. E is closed iff $E^C \cap \partial E = \emptyset$, i.e. $\partial E \subseteq E$
 $\Downarrow$
 $E^C \text{ open} \implies E^C \cap \partial(E^C) = E^C \cap \partial E = \emptyset$

5. E is closed iff it contains all of its accumulation points.

Pf. 1. $\forall x \in E_1 \cap E_2, \exists r_1, r_2 > 0$, s.t. $B(x, r_1) \subseteq E_1, B(x, r_2) \subseteq E_2$

$$\implies B(x, \min\{r_1, r_2\}) \subseteq E_1 \cap E_2$$

$$\begin{aligned} 2. (E_1 \cup E_2)^C &= E_1^C \cap E_2^C \\ (\bigcap_{\alpha \in \Lambda} E_\alpha)^C &= \bigcup_{\alpha \in \Lambda} E_\alpha^C \end{aligned}$$

$$\begin{aligned} x \in \left(\bigcap_{\alpha \in \Lambda} E_\alpha\right)^C &\iff x \notin \bigcap_{\alpha \in \Lambda} E_\alpha \\ &\iff x \notin E_\alpha \text{ for some } \alpha \in \Lambda \\ &\iff x \in E_\alpha^C \text{ for some } \alpha \in \Lambda \text{ i.e. } x \in \bigcup_{\alpha \in \Lambda} E_\alpha^C \end{aligned}$$

3. “ $\implies$ ” E is open $\implies \forall x \in E, \exists r > 0$ s.t. $B(x, r) \subseteq E \implies x \notin \partial E \implies \partial E \subset E^C$

“ $\Leftarrow$ ” If $\partial E \cap E = \emptyset$, then $\forall x \in E \implies x \notin \partial E \xrightarrow{\text{反证}} \exists r > 0$ s.t. $B(x, r) \subseteq E \implies x$ is an interior $\implies E$ open.

5. “ $\implies$ ” E closed, $\boxed{x \text{ is an accumulation point}}$, we want to prove $x \in E$ by contradiction.

Suppose $x \notin E$, then $x \in E^C$ which is open. Then $B(x, r) \subseteq E^C$ for some $r > 0$, so x is not accumulation point of E , it's a contradiction.

Therefore, $x \in E$.

“ $\Leftarrow$ ” $\forall x \in E^C$, then x is not accumulation point by assumption.

$$\implies \exists r > 0 \text{ s.t. } B^-(x, r) \cap E = \emptyset \implies B^-(x, r) \subseteq E^C$$

$$\implies B(x, r) \subseteq E^C \implies E^C \text{ open} \implies E \text{ closed.}$$

Def (Connectivity). A subset $E \subseteq \mathbb{R}^n$ is called connected if for any nontrivial $(A, B \neq \emptyset)$ decomposition $E = \underset{\text{disjoint set}}{A \sqcup B}$, either A contains some accumulation points of B

$$x \in A \cap \overline{B} \neq \emptyset \implies x \in \overline{B} \setminus B$$

$$\implies x \in (\partial B) \cap B^C$$

$$\implies x \text{ is accumulation of } B, B \cap \overline{A} \neq \emptyset$$

Def (Path-connected 道路连通). If $\forall P, Q \in E, \exists$ continuous map

$$\gamma : [0, 1] \rightarrow \mathbb{R}^n$$

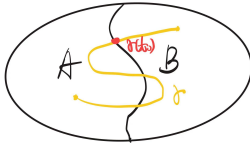
$$t \mapsto \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix}$$

s.t. $\gamma([0, 1]) \subseteq E$, and $\gamma(0) = P, \gamma(1) = Q$

Eg. $E = \{(0, y) \mid -1 \leq y \leq 1\} \cup \{(x, \sin \frac{1}{x}) \mid x \in (0, 1]\}$ **Not path connected, but it is connected**

Thm. If E is path-connected, then E is connected.

Pf. Argue by contradiction. Suppose E is not connected / disconnected. Then $A, B \neq \emptyset, A \cap B = \emptyset$, s.t. $A \sqcup B = E$ and $A \cap \overline{B} = \emptyset, B \cap \overline{A} = \emptyset$. By path-connectedness, take $x \in A, y \in B, \gamma : [0, 1] \rightarrow \mathbb{R}^n$, path connecting x and y in E to $\triangleq \sup \{t \in [0, 1] \mid \gamma(t) \in A\} \neq \emptyset$



1. $r(t_0) \in A \implies t_0 \neq 1, \forall t > t_0, r(t) \in B$
 $\implies r(t_0)$ is accumulation point of B
 $\implies r(t_0) \in \overline{B} \subseteq A^C$, contradiction.

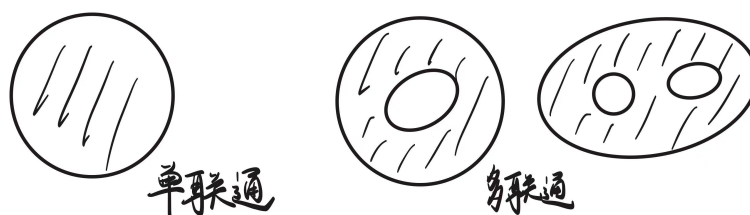
2. $r(t_0) \in B \implies \exists t_n \nearrow t_0$ s.t. $r(t_n) \in A$
 $\implies r(t_0)$ is accumulation point of A
 $\implies r(t_0) \in \overline{A} \subseteq B^C$, contradiction.

Rmk. Open set is connected iff it is path-connected.

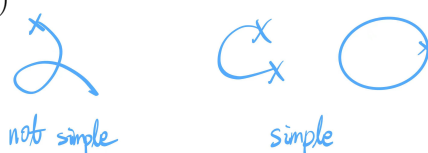
(Pick any $P \in X$, define

$$U = \{Q \in X \mid Q \text{ and } P \text{ can be connected by a path in } X\},$$

prove $U = X$)



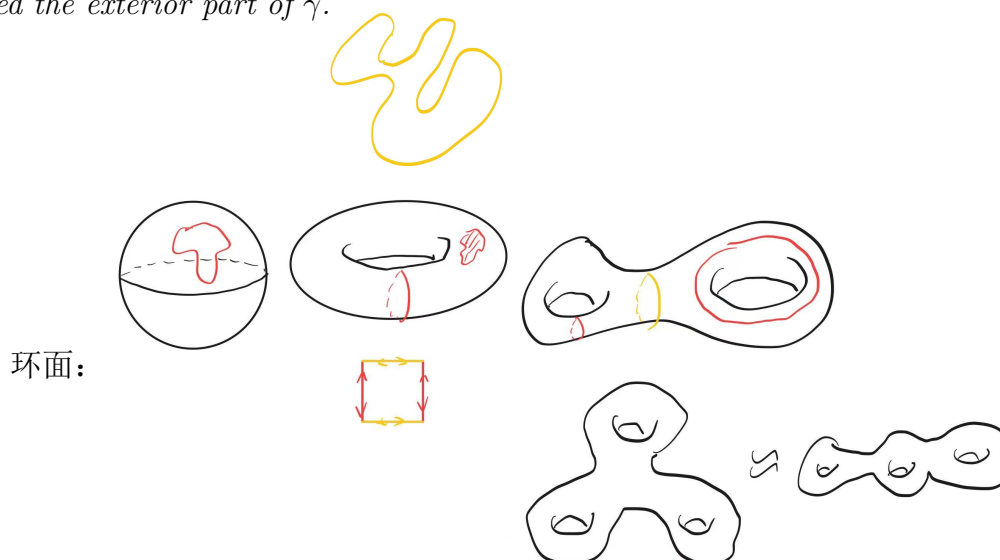
Def (Single Curve). If a curve: $\gamma : [\alpha, \beta] \rightarrow \mathbb{R}^2$ satisfies $\gamma(t_1) \neq \gamma(t_2)$ for any $\alpha \leq t_1 < t_2 \leq \beta$, then γ is called a single curve. (Not self-intersection)



Def (Loop / Closed Curve). If the curve $\gamma : [\alpha, \beta] \rightarrow \mathbb{R}^2$ has the same starting and ending point, i.e. $\gamma(\alpha) = \gamma(\beta)$, then γ is called a loop, or a closed curve.



Thm (Jordan Curve Theorem). Any simple closed curve γ on $\mathbb{R}^2$ divides $\mathbb{R}^2$ into two open connected sets, one is bounded called the interior part of γ , and the other which is unbounded called the exterior part of γ .



环面:

If the interior point of any single closed curve in D (connected open set) is contained in D , then D is called **simply-connected**. Otherwise, it is called multi-connected.

Rmk. *Topological definition (depends on the crucial notion homotopy 同伦)*

For $\mathbb{R}^2$, any loop is null-homotopic 零伦, i.e. it is homotopic to a trivial loop.

$\Pi_1(x)$ 基本群

9.1.2 Multi-variables Function

Any map

$$f : D (\subseteq \mathbb{R}^2) \rightarrow \mathbb{R}$$

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mapsto z = f(x_1, \dots, x_n)$$

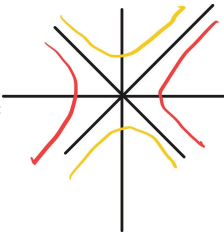
Cartesian Coordinate

is called a multi-variables function.

1. Explicit formula: $z = f(x_1, \dots, x_n)$
2. Graph: $\text{Graph}(f) = \{(x_1, \dots, x_n), f(x_1, \dots, x_n) \mid (x_1, \dots, x_n) \in D\} \subseteq \mathbb{R}^{n+1} \rightarrow$
 - $z = x^2 - y^2$ 马鞍面
 - $\{x^2 + y^2 + z^2 = 1, z \geq 0\}$ 半球: $z = \sqrt{1 - x^2 - y^2}, D = \{x^2 + y^2 \leq 1\}$
3. Level set representation

$$\forall c \subseteq \mathbb{R}, f^{-1}(c) = \{(x_1, \dots, x_n) \in D \mid f(x_1, \dots, x_n) = c\}$$

is called the level set at the level c .

Eg. • $z = f(x, y) = x^2 - y^2 \rightarrow f^{-1}(c) =$ 

• $w = f(x, y, z) = x^2 + y^2 - z^2$



Vector-valued Functions

$$f : D (\subseteq \mathbb{R}^n) \rightarrow \mathbb{R}^m$$

$$(x_1, \dots, x_n) \mapsto (y_1, \dots, y_m) = (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n))$$

Eg. 1. $n = 1, m = 3$ parameterized curve

$$\vec{r} : D (\subseteq \mathbb{R}) \rightarrow \mathbb{R}^3$$

$$t \mapsto (x(t), y(t), z(t))$$

2. $n = 2, m = 3$ parameterized surface in $\mathbb{R}^3$

$$\begin{aligned}\vec{r} : D (\subseteq \mathbb{R}^2) &\rightarrow \mathbb{R}^3 \\ (u, v) &\mapsto (x(u, v), y(u, v), z(u, v))\end{aligned}$$

Eg.

$$\begin{aligned}\vec{r} : [0, \pi] \times [0, 2\pi] &\rightarrow \mathbb{R}^3 \\ (\theta, \varphi) &\mapsto (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)\end{aligned}$$

is the spherical coordinate parameterization of the $S^2 = \partial B(0, 1)$

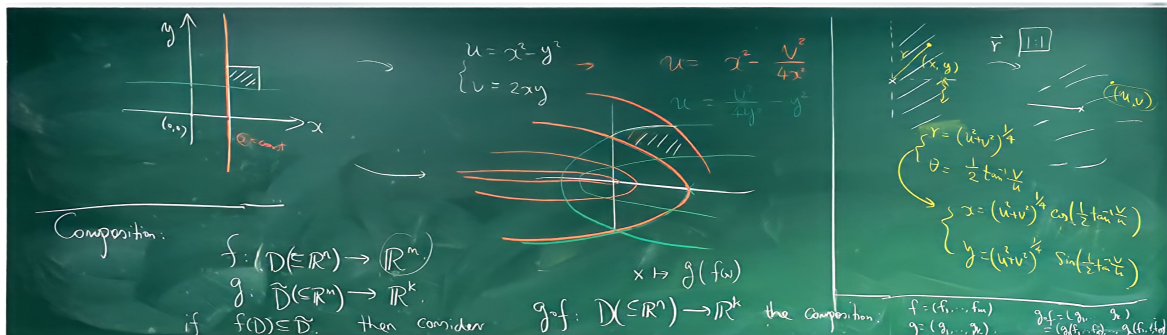
3. $n = m$ a one-to-one vector-valued function $f : D (\subseteq \mathbb{R}^b) \rightarrow \tilde{D} (\subseteq \mathbb{R}^b)$ is called a coordinate.

Transformation

Eg.

$$\begin{aligned}\vec{r} : \mathbb{R}^2 &\rightarrow \mathbb{R}^2 \\ (x, y) &\mapsto (x^2 - y^2, 2xy) \\ (r, \theta) &= (r^2 \cos 2\theta, r^2 \sin 2\theta) \rightarrow (r^2, 2\theta)\end{aligned}$$

$$\begin{aligned}x + iy &= z \\ w = u + iv &= z^2 \\ \text{Conformal 共形}\end{aligned}$$



9.1.3 Limit

Def (Limit). Let $f : D (\subseteq \mathbb{R}^2) \rightarrow \mathbb{R}$, M_0 is an accumulation point of D . If for some $a \in \mathbb{R}$, $\forall \varepsilon > 0$, $\exists \delta = \delta(\varepsilon) > 0$ s.t. $\forall M \in D \cap B^-(M_0, \delta)$, $|f(M) - a| < \varepsilon$, then we say f converges to a as M converges to M_0 , simply denote it by $\lim_{M \rightarrow M_0} f(M) = a$.

$\forall \varepsilon > 0$, $\exists \delta = \delta(\varepsilon) > 0$, s.t. $\forall (x, y) \in D$, $(x, y) \neq M_0 = (x_0, y_0)$, $|x - x_0| < \delta$, $|y - y_0| < \delta$, $|f(x, y) - a| < \varepsilon$

$$\left(\begin{array}{c} \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x, y) \\ \downarrow \\ \lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} f(x, y) \text{ 累次极限} \\ \lim_{M_0 + tv} f(M_0 + tv) \end{array} \right)$$

Similarly, we can define $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} f(x, y)$, $\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow \infty}} f(x, y)$, $\dots$

Eg. 1. Prove $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + y^2} = 0$

Pf. $\left| \frac{x^2 y^2}{x^4 + y^2} \right| = |y| \left| \frac{x^2 y}{x^4 + y^2} \right| \leq \frac{1}{2} |y|$

$\forall \varepsilon > 0$, $\exists \delta = \varepsilon$ s.t. $\forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$, $|x| < \delta$, $|y| < \delta$

$$\left| \frac{x^2 y^2}{x^4 + y^2} \right| \leq \frac{1}{2} |y| < \varepsilon$$

Q.E.D.

2. Prove $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 + y^3}{x^2 + y^2} = 0$ ($\left| \frac{x^3 + y^3}{x^2 + y^2} = 0 \right| \leq |x| + |y|$)

3. Consider $f(x, y) = \frac{x^2 y}{x^4 + y^2}$, $(x, y) \neq (0, 0)$

$$\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} \frac{x^2 y}{x^4 + y^2} = 0 \quad \lim_{x \rightarrow 0} \lim_{y \rightarrow 0} \frac{x^2 y}{x^4 + y^2} = 0$$

$$f(x, y) = f(x, kx) = \frac{x^2 \cdot kx}{x^4 + k^2 x^2} = \frac{k}{k^2 + x^2} x \rightarrow 0, \text{ as } x \rightarrow 0$$

If we consider $y = kx^2$, $f(x, y) = f(x, kx^2) = \frac{x^2 \cdot kx^2}{x^4 + k^2 x^4} = \frac{k}{1 + k^2} \rightarrow \frac{k}{1 + k^2} (x \rightarrow 0)$

$\implies \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$ does not exist

Thm. Suppose f is defined on $B^-(M_0, r)$, then $\lim_{M \rightarrow M_0} f(M) = a \iff$ for any continuous curve $\gamma : (0, \delta) \rightarrow B^-(M_0, r)$ with $\lim_{t \rightarrow 0^+} \gamma(t) = M_0$, we have $\lim_{t \rightarrow 0} f(\gamma(t)) = a$

Pf. “ $\implies$ ” $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. $\forall M \in B^-(M_0, \delta)$, $|f(M) - a| < \varepsilon$. Then for any $\gamma : (0, \delta) \rightarrow B^-(M_0, r)$ with $\gamma(t) \rightarrow M_0 (t \rightarrow 0)$. $\exists \eta > 0$ s.t. $\forall 0 < t < \eta$, $\gamma(t) \in B^-(M_0, \delta)$, then $|f(\gamma(t))| < \varepsilon$

“ $\impliedby$ ” Argue by contradiction. Suppose $\lim_{M \rightarrow M_0} f(M) \neq a$, then $\exists \varepsilon_0 > 0$ and $\{M_n\} \subseteq B^-(M_0, r)$ s.t. $M_n \rightarrow M_0 (n \rightarrow +\infty)$, and $|f(M_n) - a| \geq \varepsilon_0$

Construct a $\gamma : (0, 1) \rightarrow B^-(M_0, r)$, $\gamma(\frac{1}{2^n})$ and $\gamma(t) \rightarrow M_0 (t \rightarrow 0)$, but $f(\gamma(\frac{1}{2^n})) = f(M_n) \not\rightarrow a$. Contradiction.

Prop (Algebraic Rules). If $\lim_{M \rightarrow M_0} f(M)$, $\lim_{M \rightarrow M_0} g(M)$ both exists, then $\lim_{M \rightarrow M_0} (f \pm g)(M)$, $\lim_{M \rightarrow M_0} (f \cdot g)(M)$ exists and $\lim_{M \rightarrow M_0} (\frac{f}{g})(M)$ exists provided $\lim_{M \rightarrow M_0} g(M) \neq 0$.

Prop (Heine Principle). Suppose M_0 is an accumulation point of D , then $\lim_{M \rightarrow M_0} f(M) = a$ iff for any point sequence $\{M_n\} \subseteq D \setminus \{M_0\}$, such that $M_n \rightarrow M_0$ ($n \rightarrow +\infty$), we have $f(M_n) \rightarrow a$ ($n \rightarrow +\infty$)

$\lim_{M \rightarrow M_0} f(M)$ exists $\iff \lim_{\delta \rightarrow 0} \sup_{B^-(M_0, \delta)} |f(M) - f(M')| = 0 \iff$ In particular, $\lim_{\delta \rightarrow 0} \sup_{\partial B(M_0, \delta)} |f(M) - f(M')| = 0$

Eg. $f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$



1. Along the ray $y = kx$, $f(x, kx) = \frac{k}{1+k^2} \rightarrow \frac{k}{1+k^2}$ ($x \rightarrow 0$)

$\implies$ The limits are different along different rays

$\implies \lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist

2.

$$\begin{aligned} \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} &\implies f(\delta \cos \theta, \delta \sin \theta) = \cos \theta \sin \theta \\ &\implies \lim_{\delta \rightarrow 0} \sup_{M, M' \in \partial B(0, \delta)} |f(M) - f(M')| = 1 \\ &\implies \lim_{(x,y) \rightarrow 0} f(x, y) \text{ does not exist} \end{aligned}$$

9.1.4 Continuity

Def. Let $f : D (\subseteq \mathbb{R}^2) \rightarrow \mathbb{R}$ and $M_0 \in D$, we say f is continuous at M_0 if $\lim_{M \rightarrow M_0} f(M) = f(M_0)$, i.e. $\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0$ s.t. $\forall M \in D \cap B(M_0, \delta), |f(M) - f(M_0)| < \varepsilon$

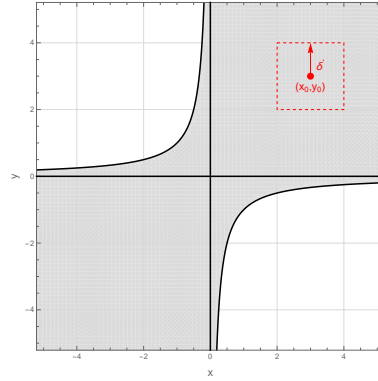
Eg. $D = \{(x, y) \in \mathbb{R}^2 \mid xy > -1\}$, $f(x, y) = \begin{cases} \log(1 + xy)^{\frac{1}{xy}} & xy \neq 0 \\ 1 & xy = 0 \end{cases}$

$$\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0, \forall |u - x_0 y_0| < \delta$$

$$\text{Need } |xy - x_0 y_0| < \delta \implies |f(x, y) - f(x_0, y_0)| < \varepsilon$$

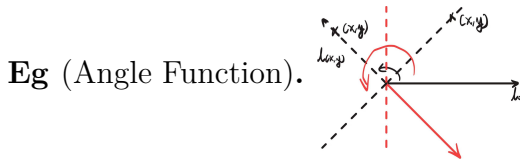
$$|xy - x_0 y_0| = |x_0| |y - y_0| - |y| |x - x_0|$$

$$\begin{aligned} (|x_0| + 2|y_0| + 1)(|x - x_0| + |y - y_0|) &\geq |x_0| |y - y_0| - |y| |x - x_0| \\ &\geq |x_0 y - x_0 y_0| + |xy - x_0 y| \\ &\geq |f(x, y) - f(x_0, y_0)| \end{aligned}$$



$$\text{Take } \delta' = \min \left\{ \frac{\delta}{2(|x_0| + 2|y_0| + 1)} |y_0| \right\}.$$

$$\text{Then } \forall |x - x_0| < \delta', |y - y_0| < \delta', |xy - x_0 y_0| < \delta', |f(x, y) - f(x_0, y_0)| < \varepsilon$$



$$\begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \text{the angle between the ray } l_0 \text{ and } l_{(x,y)} \text{ counted anti-clockwise} \end{cases}$$

$$(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$$

$$= \begin{cases} \arctan \frac{y}{x} & x > 0, y > 0 \\ \frac{\pi}{2} & x = 0, y > 0 \\ \pi + \arctan \frac{y}{x} & x < 0, y > 0 \\ \frac{3\pi}{2} & x = 0, y < 0 \\ 2\pi + \arctan \frac{y}{x} & x > 0, y < 0 \end{cases}$$

$$\begin{aligned} \tan : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) &\xrightarrow{1:1} \mathbb{R} \\ \xrightarrow{\text{inverse}} \arctan : \mathbb{R} &\rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \end{aligned}$$

Conclusion:

1. θ is continuous on $\mathbb{R}^2 \setminus [0, +\infty)$ but has jumping discontinuity on $(0, +\infty)$
2. r is continuous on $\mathbb{R}^2$

Def (Uniform Continuity). Let f be defined on $D \subseteq \mathbb{R}^2$, f is said to be uniformly continuous if $\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0$, s.t. $\forall M', M'' \in D, d(M', M'') < \delta, |f(M') - f(M'')| < \varepsilon$

Thm. 1. The continuity of function is preserved under $+, -, \times$ and $\div$ (provided the denominator function is nonzero at the point of interest)

2. The composition of continuous vector-valued functions is continuous.

3. (Intermediate Value Property) Let f be defined on a connected set $D \subseteq \mathbb{R}^2$, and f is continuous. Then for any $M_1, M_2 \in D$, all values between $f(M_1)$ and $f(M_2)$ can be achieved. In particular, $f(D)$ is an interval. (More generally, the image of a connected set under a continuous vector-valued function is connected.)

Pf. Argue by contradiction.

Suppose $f(D)$ is not connected. Then $f(D) = A \sqcup B$, $A, B \neq \emptyset$, $A \cap \overline{B} = \emptyset$, $B \cap \overline{A} = \emptyset$.

$$\implies D = f^{-1}(A) \sqcup f^{-1}(B), f^{-1}(A), f^{-1}(B) \neq \emptyset$$

Suppose $\{M_n\} \subseteq f^{-1}(A)$, $M_n \rightarrow M_0 \in f^{-1}(B)$, then $f(\overset{A}{\downarrow} M_n) \rightarrow f(\overset{B}{\downarrow} M_0) \implies B \cap \overline{A} \neq \emptyset$, contradiction.

$$\implies \overline{f^{-1}(A)} \cap f^{-1}(B) = \emptyset$$

Similarly, $\implies \overline{f^{-1}(B)} \cap f^{-1}(A) = \emptyset$

$\implies$ This is a contradiction with the connectivity of D .

4. The $\overset{\text{supremum}}{\uparrow}$ maximum and $\overset{\text{infimum}}{\uparrow}$ minimum of a continuous function on a bounded closed set can be achieved.

Pf. Step 1) $\sup_D f < +\infty, \inf_D f > -\infty$

If $\sup_D f = +\infty$, then $\exists \{M_n\} \subseteq \overset{\text{closed, bounded}}{\downarrow} D$, s.t. $f(M_n) \rightarrow +\infty$

By Bolzano-Weierstrass Thm, $\exists$ subsequence $\{M_{n_k}\}$ s.t. $M_{n_k} \rightarrow M_0 \in D (k \rightarrow +\infty)$, $f(M_{n_k}) \rightarrow f(M_0) = +\infty (k \rightarrow +\infty)$ by continuity, contradiction.

Step 2) $\sup_D f = f(M_0)$ for some $M_0 \in D$

Suppose $\sup_D f \neq f(M)$ for any $M \in D$

$$\implies \exists \{M_n\} \subseteq D \text{ s.t. } f(M_0) \rightarrow \sup_D f$$

$$\stackrel{B.W.}{\implies} \exists \text{ subsequence } \{M_{n_k}\} \subseteq D, M_{n_k} \rightarrow M_0 \in D \implies f(M_{n_k}) \rightarrow f(M_0) = \sup_D f$$

5. (Cantor Thm) *Continuous function on a bounded closed set is uniformly continuous.*

9.2 多变量函数的微分

9.2.1 Differentiation

$$\int_b^a df = \int_{\partial[a,b]} f$$

$$\begin{aligned}\frac{\partial f}{\partial x}(x_0, y_0) &= \lim_{x \rightarrow x_0} \frac{f(x, y_0) - f(x_0, y_0)}{x - x_0} \\ \frac{\partial f}{\partial y}(x_0, y_0) &= \lim_{y \rightarrow y_0} \frac{f(x_0, y) - f(x_0, y_0)}{y - y_0}\end{aligned} \quad (\text{assume domain is open set})$$

are called the first order partial derivatives of f at (x_0, y_0) .

If $\frac{\partial f}{\partial x}(x, y)$ exists, for any $(x, y) \in D$, then the function $\frac{\partial f}{\partial x}$ is called the first order partial derivative (function), also denoted by f'_x, f'_y

Similarly, we can define

$$\begin{aligned}\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) &= \frac{\partial^2 f}{\partial x^2} = f''_{xx} & \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) &= \frac{\partial^2 f}{\partial y^2} = f''_{yy} \\ \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) &= \frac{\partial^2 f}{\partial y \partial x} = f''_{xy} \quad \text{not necessarily} \neq \quad \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} = f''_{yx}\end{aligned}$$

mixed 2nd-order partial derivatives

Geometric Meaning of Second-order Partial Derivatives

$$\begin{aligned}\left. \frac{\partial^2 f}{\partial x^2} \right|_{(x_0, y_0)} &= \left. \frac{d}{dx} \right|_{x=x_0} \frac{\partial f}{\partial x}(x, y_0) = \left. \frac{d}{dx} \right|_{x=x_0} \left. \frac{d}{dt} \right|_{t=x} f(t, y_0) \\ &= \lim_{\Delta x \rightarrow 0} \frac{\frac{\partial f}{\partial x}(x_0 + \Delta x, y_0) - \frac{\partial f}{\partial x}(x_0, y_0)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \lim_{\Delta t \rightarrow 0} \frac{f(x_0 + \Delta x + \Delta t, y_0) - f(x_0 + \Delta x, y_0) - f(x_0 + \Delta t, y_0) + f(x_0, y_0)}{\Delta t} \\ &= \lim_{\Delta x \rightarrow 0} \lim_{\Delta t \rightarrow 0} \frac{f(x_0 + \Delta x + \Delta t, y_0) - f(x_0 + \Delta x, y_0) - f(x_0 + \Delta t, y_0) + f(x_0, y_0)}{\Delta x \Delta t}\end{aligned}$$

↓
The second order ordinary derivative of the function $f(x, y_0)$ at x_0

$$\begin{aligned}\left. \frac{\partial^2 f}{\partial y \partial x} \right|_{(x_0, y_0)} &= \lim_{\Delta y \rightarrow 0} \frac{\frac{\partial f}{\partial x}(x_0, y_0 + \Delta y) - \frac{\partial f}{\partial x}(x_0, y_0)}{\Delta y} \\ &= \lim_{\Delta y \rightarrow 0} \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y) - f(x_0 + \Delta x, y_0) + f(x_0, y_0)}{\Delta x \Delta y}\end{aligned}$$

↓
 $\frac{\Pi(\Delta x, \Delta y)}{\Delta x \Delta y}$

$$\left. \frac{\partial^2 f}{\partial x \partial y} \right|_{(x_0, y_0)} = \lim_{\Delta x \rightarrow 0} \lim_{\Delta y \rightarrow 0} \frac{\Pi(\Delta x, \Delta y)}{\Delta x \Delta y}$$

Thm. If $\frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}$ exist in an open neighborhood of (x_0, y_0) and are both continuous at (x_0, y_0) , then $\left. \frac{\partial^2 f}{\partial y \partial x} \right|_{(x_0, y_0)} = \left. \frac{\partial^2 f}{\partial x \partial y} \right|_{(x_0, y_0)}$

Pf.

$$\begin{aligned}
& f(x_0 + h, y_0 + k) - f(x_0, y_0 + k) - f(x_0 + h, y_0) + f(x_0, y_0) \\
&= \phi(h) - \phi(0) \text{ where } \phi(h) = f(x_0 + h, y_0 + k) - f(x_0 + h, y_0) \\
&\quad \phi(0) = f(x_0, y_0 + k) - f(x_0, y_0) \\
&\stackrel{\exists \theta \in (0,1)}{\underset{L.M.V.T.}{=}} \phi'(\theta h)h = \left(\frac{\partial f}{\partial x}(x_0 + \theta h, y_0 + k) - \frac{\partial f}{\partial x}(x_0 + \theta h, y_0) \right) h \\
&\stackrel{\exists \eta \in (0,1)}{\underset{L.M.V.T.}{=}} \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \Big|_{(x_0 + \theta h, y_0 + \eta k)} kh
\end{aligned}$$

$$\text{Similarly, } \Pi(h, k) = \frac{\partial^2 f}{\partial x \partial y}(x_0 + \theta h, y_0 + \eta k)hk = \frac{\partial^2 f}{\partial y \partial x}(x_0 + \theta h, y_0 + \eta k)hk$$

$$\text{Take limit } h, k \rightarrow 0 \quad \frac{\partial^2 f}{\partial x \partial y} \Big|_{(x_0, y_0)} = \frac{\partial^2 f}{\partial y \partial x} \Big|_{(x_0, y_0)}$$

Thm. If $f''_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)$ and $\frac{\partial^2 f}{\partial y \partial x}$ are both continuous at $M_0 = (x_0, y_0)$, then

$$\frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) = \frac{\partial^2 f}{\partial y \partial x}(x_0, y_0)$$

Eg. $\boxed{f(x) = x^2 \sin \frac{1}{x} \quad f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x} \quad \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = 0}$

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}(x^2 - y^2) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

(1) f is continuous on $\mathbb{R}^2$

(2)

$$\begin{aligned}
\frac{\partial f}{\partial x}(x, y) &= \frac{y(x^2 - y^2) + 2x^2y}{x^2 + y^2} - \frac{2x^2y(x^2 - y^2)}{(x^2 + y^2)^2}, \quad (x, y) \neq (0, 0) \\
&= y \frac{(3x^2 - y^2)(x^2 + y^2) - 2x^2(x^2 - y^2)}{(x^2 + y^2)^2} \\
&= y \frac{x^4 + 4x^2y^2 - y^4}{(x^2 + y^2)^2}
\end{aligned}$$

$\frac{\partial f}{\partial x}(0, 0) = 0$ since $\lim_{x \rightarrow 0} \left(\frac{\partial f}{\partial x}(x, 0) \right) = \lim_{x \rightarrow 0} \frac{d}{dx}(f(x, 0)) = \lim_{x \rightarrow 0} \frac{d}{dx}(0) = 0$, then using Darboux's Thm for one-variable function.

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{0 - 0}{x} = 0 \text{ (direct computation by definition)}$$

Similarly,

$$\frac{\partial f}{\partial y}(x, y) = \begin{cases} -x \cdot \frac{y^4 + 4x^2y^2 - x^4}{(x^2 + y^2)^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

In conclusion, $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ are both continuous on $\mathbb{R}^2$

$$(3) \quad \frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}, \frac{\partial^2 f}{\partial y^2}$$

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2}(x, y) &= y \cdot \frac{(4x^3 + 8xy^2)(x^2 + y^2)^2 - 4x(x^2 + y^2)(x^4 + 4x^2y^2 - y^4)}{(x^2 + y^2)^4} \\ &= 4xy \cdot \frac{(x^2 + 2y^2)(x^2 + y^2) - (x^4 + 4x^2y^2 - y^4)}{(x^2 + y^2)^3} \\ &= 4xy \cdot \frac{3y^4 - x^2y^2}{(x^2 + y^2)^3} \\ &= 4xy^3 \cdot \frac{3y^2 - x^2}{(x^2 + y^2)^3}, \quad (x, y) \neq (0, 0) \end{aligned}$$

$$\frac{\partial^2 f}{\partial x^2}(0, 0) = \begin{cases} \lim_{x \rightarrow 0} \frac{\frac{\partial f}{\partial x}(x, 0) - \frac{\partial f}{\partial x}(0, 0)}{x - 0} = 0 \\ = 0 \quad \text{by computing } \lim_{x \rightarrow 0} \frac{\partial^2 f}{\partial x^2}(x, 0) \end{cases}$$

$$\frac{\partial^2 f}{\partial y^2} \dots \text{ (omitted)}$$

$$\begin{aligned} \frac{\partial^2 f}{\partial x \partial y}(x, y) &= -\frac{\partial}{\partial x} \left(\frac{x(y^4 + 4x^2y^2 - x^4)}{(x^2 + y^2)^2} \right) \\ &= -\left(\frac{(y^4 + 4x^2y^2 - x^4) + (8xy^2 - 4x^3)x}{(x^2 + y^2)^2} - \frac{4x^2(y^4 + 4x^2y^2 - x^4)}{(x^2 + y^2)^3} \right) \\ &= -\frac{(y^4 + 12x^2y^2 - 5x^4)(x^2 + y^2) - 4x^2(y^4 + 4x^2y^2 - x^4)}{(x^2 + y^2)^3} \\ &= -\frac{y^6 + 9y^4x^2 - 9y^2x^4 - x^6}{(x^2 + y^2)^3}, \quad (x, y) \neq (0, 0) \end{aligned}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \dots = \frac{\partial^2 f}{\partial x \partial y}, \quad (x, y) \neq (0, 0)$$

= rational function of x, y with nonzero denominator around $(x, y) \neq (0, 0)$

Therefore $\frac{\partial^2 f}{\partial y \partial x}$ is continuous at $(x, y) \neq (0, 0)$

$$\implies \frac{\partial^2 f}{\partial y \partial x}(x, y) = \frac{\partial^2 f}{\partial x \partial y}(x, y) \text{ for } (x, y) \neq (0, 0)$$

$$\begin{aligned} \frac{\partial^2 f}{\partial y \partial x}(0, 0) &= \lim_{y \rightarrow 0} \frac{\frac{\partial f}{\partial x}(0, y) - \frac{\partial f}{\partial x}(0, 0)}{y} \\ &= \lim_{y \rightarrow 0} \frac{-y^4}{(y^2)^2} = -1 \\ \frac{\partial^2 f}{\partial x \partial y}(0, 0) &= \lim_{x \rightarrow 0} \frac{\frac{\partial f}{\partial y}(x, 0) - \frac{\partial f}{\partial y}(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{x^4}{(x^2)^2} = 1 \end{aligned}$$

In conclusion, $\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}, \frac{\partial^2 f}{\partial y^2}$ are all bounded, are all discontinuous at $(0, 0)$.

f is "smooth" on $\mathbb{R}^2 \setminus \{(0,0)\}$.

Furthermore, $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ are locally Lipschitz continuous, i.e. $|\frac{\partial f}{\partial x}(M_1) - \frac{\partial f}{\partial x}(M_2)| \leq Cd(M+1, M_2)$ for some $C > 0$, $M_1, M_2 \in B(0,1)$.

Rmk.

$$\begin{aligned}\Pi(h, k) &= (f(h, k) - f(h, 0)) - (f(0, k) - f(0, 0)) \\ &= (f(x, k) - f(x, 0))|_0^h \\ &= \left(\frac{\partial f}{\partial x}(\theta h, k) - \frac{\partial f}{\partial x}(\theta h, 0) \right) h \\ &= \left(\frac{\partial f}{\partial y}(h, \eta k) - \frac{\partial f}{\partial y}(0, \eta k) \right) h\end{aligned}$$

Eg. $f(x, y) = F(x) + G(y)$ where $F \in C^1$, G is arbitrary.

- $\frac{\partial f}{\partial x}(x, y) = F'(x) \implies \boxed{\frac{\partial^2 f}{\partial y \partial x}(x, y) \equiv 0}$
- $\frac{\partial f}{\partial y}$ does not necessarily exist, let alone $\frac{\partial^2 f}{\partial x \partial y}$
- If $F, G \in C^1$, $\frac{\partial^2 f}{\partial x \partial y} \equiv \frac{\partial^2 f}{\partial y \partial x} \equiv 0$, but $\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}$ does not necessarily exist.

In contrast, look at

Eg. Assume $\frac{\partial^2 f}{\partial x^2} \equiv 0$, $\frac{\partial^2 f}{\partial y^2} \equiv 0$.

- $\frac{\partial^2 f}{\partial x^2} \equiv 0 \implies f$ is a linear function of x on $\{(t, y) \mid t \in \mathbb{R}\} \implies f(x, y) = G(y) + A(y)x$, where A, G are arbitrary.
- $\frac{\partial^2 f}{\partial y^2} \equiv 0 \implies$
 - $G(y) = a + by$ where a, b are constant.
 - $A(y) = c + dy$ where c, d are constant.

$a + by + (c + dy)x = a + by + cx + dxy$, $a, b, c, d \in \mathbb{R}$ Solution.

Rmk. $\Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0 \implies f$ has some good property.

9.2.2 Differentiability

Recall: f is differentiable at x_0 iff $f(x_0 + \Delta x) - f(x_0) = a\Delta x + o(|\Delta x|)$, $\Delta x \rightarrow 0$
 ($\iff f$ has derivative at x_0 and $a = f'(x_0)$, $\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$ exist)

Def. Let $f : D(\subseteq \mathbb{R}^n) \rightarrow \mathbb{R}$, x is an interior point of D if

$$f(x + h) = f(x) + A(h) + o(|h|), \quad h \rightarrow 0$$

$\implies$ If f is differentiable at x , then the graph of f has a tangent *hyperplane* (in $\mathbb{R}^{6n+1}$) which is the graph of $\begin{smallmatrix} h \mapsto f(x) + A(h) \\ \mathbb{R}^n \rightarrow \mathbb{R} \end{smallmatrix}$

When $A : \mathbb{R}^n \rightarrow \mathbb{R}$ which is a linear function, then we say f is differentiable at x , the map A is called the differential of f at x , denoted by $df|_x$

If f is differentiable at any $x \in D$, then we say f is differentiable on D and $\{df|_x \mid x \in D\} = df$ is called the differential of f on D .

Rmk. For $n = 1$, $\{f'(x) \mid x \in D\} = f'$ is called the derivative function of f on D . We also have $df = f'(x)dx$, called the differential of f on D .

Thm. If f is differentiable at x , then it is continuous at x .

Assume f is differentiable at x , then $f(x + h) = f(x) + A(h) + o(|h|)$, $h \rightarrow 0$
 $\implies f(x + te_i) = f(x) + A(te_i) + o(|t|) = f(x) + tA(e_i) + o(|t|)$, $t \rightarrow 0$ ($i = 1, 2, \dots, n$)
 $\implies \frac{\partial f}{\partial x_i}(x)$ exists (since $\frac{\partial f}{\partial x_i}(x) = \lim_{t \rightarrow 0} \frac{f(x + te_i) - f(x)}{t}$), and $\frac{\partial f}{\partial x_i}(x) = A(e_i)$

Thm. If f is differentiable at x , then $\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x)$ all exist, and

$$\frac{\partial f}{\partial x_i}(x) = df|_x(e_i)$$

The converse is not true.

Look at some special functions: $\boxed{X_i : \mathbb{R}^n \rightarrow \mathbb{R}, i = 1, \dots, n}$ (The n Cartesian coordinates function)

$x_i(x + h) = x_i(x) + h_i$, where $h_i = i$ -th component of h under the standard basis $\{e_1, \dots, e_n\}$

$\implies x_i$ is differentiable on $\mathbb{R}^n$, so we have dx_i

$$\begin{aligned}
f(x+h) &= f(x) + df|_x(h) + o(|h|) \\
&= f(x) + df|_x \left(\sum_{i=1}^n h_i e_i \right) + o(|h|) \\
&= f(x) + \sum_{i=1}^n df|_x(e_i) h_i + o(|h|) \\
&= f(x) + \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x) h_i + o(|h|) \\
&= f(x) + \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x) dx^i(h) + o(|h|)
\end{aligned}$$

$f(x) + \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x) h_i + o(|h|) \rightarrow d\mathbf{f}|_x = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(\mathbf{x}) d\mathbf{x}_i$ “The linear *combination* of the n special linear function $dx_1, \dots, dx_n$, with coefficients the partial derivatives.”

$dx_1, \dots, dx_n$ are the standard differential 1-form.

Further, we can write

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i$$

“module” 模

$$\left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)$$

$\text{Grad } f|_x = \nabla f|_x = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x) e_i$ is called the gradient 梯度 of f at x .

9.2.3 方向导数与梯度

9.2.4 复合函数的微分和一阶微分形式不变性

9.2.5 向量值函数的微商和微分

9.3 隐函数定理和逆映射定理

9.3.1 隐函数的存在性和微商

9.3.2 从微分的角度看隐函数定理

9.3.3 逆映射的微商

9.4 空间曲线与曲面

9.4.1 参数曲线

9.4.2 参数曲面

9.4.3 隐式曲线和隐式曲面

9.5 多变量函数的 Taylor 公式与极值

9.5.1 二元函数的微分中值定理

9.5.2 二元函数的 Taylor 公式

笔记

9.5.3 二元函数的极值

笔记

9.5.4 条件极值

笔记

9.6 向量场的微商

9.6.1 向量场

9.6.2 梯度、散度与旋度

9.6.3 Hamilton 算子在柱面坐标系和球面坐标系中的表示

9.7 微分形式

9.7.1 微分形式的空间

9.7.2 微分形式的外积

9.7.3 微分形式的外微分

9.7.4 微分形式在高维空间的推广

10 多变量函数的重积分

10.1 二重积分

10.1.1 平面区域的面积

10.1.2 二重积分的基本概念与性质

10.1.3 二重积分的计算

10.2 二重积分的换元

10.2.1 坐标曲线和面积元素

10.2.2 二重积分的换元

10.3 三重积分

10.3.1 三重积分的累次积分

10.3.2 三重积分的换元

10.3.3 来自物理学中的几个例子

10.4 n 重积分

11 曲线积分和曲面积分

11.1 数量场在曲线上的积分

11.1.1 基本概念

11.1.2 数量场在曲线上的积分的计算

11.2 数量场在曲面上的积分

11.2.1 曲面的面积

11.2.2 数量场在曲面上的积分的计算

11.3 向量场在曲线上的积分

11.3.1 曲线的定向

11.3.2 向量场在曲线上的积分的定义和计算

11.3.3 Green 定理

11.4 向量场在曲面上的积分

11.4.1 双侧曲面及其定向

11.4.2 向量场在曲面上的积分的定义和计算

11.5 Gauss 定理和 Stokes 定理

11.5.1 Gauss 定理

11.5.2 Stokes 定理

11.6 其他形式的曲线、曲面积分

11.7 保守场

11.7.1 保守场与势函数

11.7.2 无源场与向量势

11.8 微分形式的积分

11.8.1 微分形式的积分

11.8.2 全微分方程

12 Fourier 分析

12.1 函数的 Fourier 级数

12.1.1 周期函数与三角函数的正交性

12.1.2 周期函数的 Fourier 级数

12.1.3 有限区间上函数的 Fourier 级数

12.1.4 Fourier 级数的复数形式

12.2 平方平均收敛

12.2.1 基本概念

12.2.2 Bessel (贝塞尔) 不等式

12.2.3 平方平均收敛

12.2.4 广义 Fourier 级数

12.3 收敛性定理的证明

12.3.1 Dirichlet 定理的证明

12.3.2 平方平均收敛性定理的证明

12.4 Fourier 变换

12.4.1 Fourier 积分

12.4.2 Fourier 变换

13 反常积分和含参变量的积分

13.1 反常积分

13.1.1 无穷区间上积分的收敛性

13.1.2 无穷区间上积分收敛性的一般判别法

13.1.3 无界函数积分的收敛判别法

13.2 反常多重积分

13.3 含参变量的积分

13.3.1 含参变量的积分及其性质

13.3.2 积分限依赖于参变量的积分及其性质

13.4 含参变量的反常积分

13.4.1 含参变量的反常积分的一致收敛性

13.4.2 含参变量反常积分的性质

13.4.3 几个重要的积分

13.5 Euler 积分

13.5.1 Γ 函数的性质

13.5.2 B 函数的性质

Def (定义标题). 定义内容...

Thm (定理标题). 定理内容...

证明. 证明过程...

□

Eg (例题标题). 例题内容...

Rmk. 备注说明...

13.6 重要公式

...

(1)

13.7 课后总结

14 第 $X+1$ 章：[章节标题]